

POCKETSMART AI

03_Project_Design

Project Design Document

Phase 3 – Project Design

This document presents the system architecture, use case design, data flow, application flow, database design, and UI wireframe plan for PocketSmart AI.

1. System Architecture Diagram

PocketSmart AI uses a responsive web frontend connected to a FastAPI backend. The backend manages authentication, planner requests, recommendation generation, history, database operations, and external/AI recommendation services.

User	Frontend	FastAPI Backend	Services / AI	Database
Browser / Mobile	HTML + CSS + JavaScript	Routes + Authentication + Planner APIs	Gemini + Fallback + Catalog	SQLite
Register / Login	Pages & forms	JWT authentication	Structured recommendation generation	Users
Planner input	Planner UI	Planner routes	Home / Party / Jewelry logic	Recommendations
View history	History UI	History routes	Recommendation history	Recommendation records

Architecture flow: User → Frontend → FastAPI API → Planner/Recommendation Services → Gemini or Fallback Engine → Catalog Data → SQLite → Frontend.

2. Use Case Diagram

The main actor is the User. The user can register, log in, use planners, receive recommendations, view details, view recommendation history, and log out.

User	System Use Cases
User	Register / Login
User	Logout
User	Home Interior Planner
User	Party Planner
User	Jewelry Planner
User	Upload optional outfit image
User	Generate AI recommendations
User	View recommendation details
User	View recommendation history

Use-case relationships: Authentication is required before protected planner/history operations. Planner requests invoke recommendation generation. Generated results can be stored and later retrieved through History.

3. Data Flow Diagram (DFD)

Level 0 – Context Flow

External Entity	Process	Data Store / Output
User	PocketSmart AI	SQLite Database
User input: account / planner requirements	Authentication + Planner + Recommendation processing	User and recommendation records
Optional outfit image	AI / fallback recommendation engine	Recommendation history
Requested recommendations	Result presentation	Recommendation details

Level 1 – Main Data Flow

Step	Input	Processing	Output
1	Registration/Login data	Validate credentials and create JWT session	Authenticated user
2	Planner requirements	Validate planner request	Planner data
3	Planner data	Gemini structured output or fallback engine	Recommendation set
4	Recommendation set	Catalog matching / formatting	Display-ready recommendations
5	Generated result	Save recommendation record	History entry
6	History request	Read saved records	History/details page

4. Application Flowchart

The following sequence describes the normal user flow through PocketSmart AI.

Flow	Action
START	Open PocketSmart AI
↓	Register or Login
↓	Authentication successful?
YES	Open application dashboard / planner area
↓	Choose Home Interior, Party, or Jewelry Planner
↓	Enter requirements; optionally upload outfit image where supported
↓	Submit planner request
↓	Generate recommendations using Gemini; use fallback engine when required
↓	Match/format catalog recommendations
↓	Display recommendations and details
↓	Save result to recommendation history
↓	User may view History / Details
↓	Logout
END	Session terminated

5. ER Diagram / Database Design

The project uses SQLite for persistent storage. The main database entities are User and Recommendation. A user can have multiple recommendation records.

Entity	Important Fields	Relationship
User	id, username/email, password hash, created information	One User → Many Recommendations
Recommendation	id, user_id, planner type, input/request data, generated result, timestamp	Belongs to one User

Relationship: User (1) ■■■■■■■■■■ (M) Recommendation

Database design notes: user_id in Recommendation links each saved recommendation to its owner. Authentication data and recommendation history are kept separately so the application can retrieve a user's previous planner results.

6. UI Design / Wireframes

The frontend is designed as responsive HTML/CSS/JavaScript pages. The planned screens correspond to the project templates.

Screen	Main UI Elements	Purpose
Authentication	Login / Register form, validation messages	Account access
Home / Dashboard	Navigation, planner options, main actions	Start a planning task
Home Interior Planner	Requirement fields, submit button, result area	Generate interior recommendations
Party Planner	Party requirement fields, submit button, results	Generate party recommendations
Jewelry Planner	Jewelry requirement fields, optional outfit image, results	Generate jewelry recommendations
Details	Recommendation information and relevant catalog details	View selected result
History	Previous recommendation entries	Review earlier results

UI design principles: simple navigation, responsive layout, clear planner forms, readable recommendation cards, and easy access to history/details.

7. Code Files Supporting the Design

The uploaded Phase 3 format specifies that the design should be supported by the project's code files. For PocketSmart AI, the relevant structure includes:

Design Area	Supporting Project Files
Database / Models	app/db.py, app/models/user.py, app/models/recommendation.py, app/models/schemas.py
Backend / APIs	app/main.py, app/routes/pages.py, app/routes/auth.py, app/routes/planners.py, app/routes/history.py
Authentication	app/services/auth.py
AI / Recommendation	app/services/gemini_utils.py, app/services/fallback.py, app/services/catalog.py
Frontend Pages	templates/base.html, templates/auth.html, templates/index.html, templates/planner.html, templates/details.html, templates/history.html
Frontend Styling / Scripts	static/css/styles.css, static/js/app.js

Conclusion

This Project Design document defines how PocketSmart AI is organized from user interaction to frontend, backend, AI/recommendation services, and database storage. It also provides the design basis for implementation and testing in the later project phases.